

CIE

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TITLE: **Expense Manager**

SCHOOL: Chitkara University, Rajpura, Punjab

**DETAIL OF THE STUDENT/MENTOR WHO UPLOADED THE COPYRIGHT ON GOVT PORTAL NEED TO
SHARE LOGIN AND PASSWORD WITH MENTORS**

Name: Shivam Gupta

Roll No/ Employee Code: 2210992309

ALL MEMBERS DETAILS INCLUDING SUPERVISOR

NAME	EMP CODE/ROLL	ADDRESS WITH EMAIL AND MOBILE NO.
Student 1: Shivam Gupta	2210992309	ADDRESS : Jawahar Nagar, Moga , Punjab. Email ID : shivam2309.be22@chitkara.edu.in . Mobile No. 9888803901
Student 2: Kashish	2210991764	ADDRESS : Janta Colony, Rampura , Punjab. Email ID : kashish1764.be22@chitkara.edu.in . Mobile No. 6239522330
Student 3		
Student 4		
Mentor :Dr. Shikha Tutneja	Assistant Professor	Email ID : shikha.1290@chitkara.edu.in . Mobile No. 9915369888

ABSTRACT

1.1 Problem Background

Managing daily financial expenses has become increasingly challenging in today's fast-paced digital environment. Many individuals, especially students and working professionals, struggle to maintain accurate and organized records of their income and expenditures. Traditional methods such as notebooks or spreadsheets are time-consuming, lack automation, and do not provide real-time insights or analytical support.

Additionally, the absence of proper tracking often leads to overspending, poor budgeting decisions, and lack of financial awareness. There is a strong need for a smart, automated, and user-friendly system that not only records expenses but also provides real-time analysis, categorized tracking, and visual insights to support better financial planning and decision-making.

1.2 Objective of the Project

The objective of this project is to design and develop a web-based Expense Manager Application using modern web technologies that allows users to efficiently track, manage, and analyze their daily expenses.

The system aims to provide categorized expense tracking, real-time budget monitoring, and graphical visualization of financial data. It also focuses on improving user experience through an interactive interface, secure authentication, and efficient data handling.

1.3 Proposed Solution / System Overview

The proposed Expense Manager Application is a full-stack web-based system developed using the MERN stack (MongoDB, Express.js, React.js, and Node.js). The application provides a secure platform with user authentication features such as login, signup, and logout.

The system includes a dashboard that displays total expenses, average monthly spending, and budget usage. Users can add expenses through a structured form, and the data is stored in a MongoDB database via RESTful APIs. The application also provides graphical representations such as bar charts and pie charts for better analysis of monthly expenses.

The system follows a client-server architecture where the frontend communicates with the backend through APIs, ensuring efficient data processing and real-time updates.

1.4 Key Features or Functionality

The application includes several key features such as light and dark mode switching, categorized expense tracking (Food, Travel, Health, Fitness, Shopping, Bills, Others), and real-time budget monitoring with alerts when spending exceeds predefined limits.

Additional features include graphical data visualization using bar charts and pie charts, search and filter functionality for expenses, CSV export & with bulk import option, and detailed transaction history with date-wise records. The system ensures a smooth and user-friendly experience with responsive design and efficient performance.

1.5 Technology / Method Used

The Expense Manager Application is developed using the MERN stack, which includes MongoDB, Express.js, React.js, and Node.js. MongoDB is used as a NoSQL database to store user and expense data efficiently. Express.js and Node.js are used for backend development to handle server-side logic and API creation. React.js is used to build a dynamic and interactive user interface.

The application follows a client-server architecture where the frontend communicates with the backend through RESTful APIs. This ensures smooth data flow, real-time updates, and efficient performance.

The complete source code of the project is maintained on GitHub and can be accessed through the following link: <https://github.com/Shivam1514/Expense-Manager>

The repository includes the full implementation of the application, including frontend components, backend APIs, database models, and configuration files, ensuring proper version control and ease of future enhancements.

1.6 Expected Outcome & Benefits

The system is expected to improve financial awareness and help users maintain better control over their expenses. It provides real-time insights into spending patterns, enabling users to make informed financial decisions.

The application reduces manual effort, minimizes errors, and enhances budgeting discipline. Graphical analysis and categorized tracking help users understand their financial behavior, while features like export and filtering provide flexibility for long-term expense management.

1.7 Original Contribution Statement

This project presents a user-friendly and efficient expense management system with real-time budget tracking, interactive data visualization, and secure authentication. The integration of modern web technologies and intuitive design makes it particularly useful for students and individuals seeking better financial management.

The project also demonstrates practical implementation of full-stack development using the MERN stack, making it scalable and adaptable for future enhancements.